

Use AOP MCP tool and look into KE1977, KE2116 and KE2107 .

Can they be connected through new Key Event Relationships? Why or why not? Is it possible to use existing common relationships and KE upstream ?

Analysis: KE1977, KE2116, and KE2107 — Connectivity Assessment

What Each KE Represents

KE	Name	Biological Level	Network Status
KE1977	Disruption of sodium channel gating kinetics	Cellular	Well-connected
KE2116	Decreased oligodendrocyte differentiation	Cellular	Orphan — no upstream or downstream KERs
KE2107	Hypomyelination	Tissue	Partially connected

Can They Be Connected Through New KERs?

Yes — and there is strong biological rationale to do so. Here is the case for each proposed linkage:

1. KE2116 → KE2107 (New KER: Decreased oligodendrocyte differentiation → Hypomyelination)

This is the most defensible new KER of the three. Oligodendrocytes are the cells responsible for producing myelin in the CNS. If their differentiation from precursor cells (OPCs) is impaired, fewer mature myelinating oligodendrocytes arise, directly reducing myelin sheath formation. This is a well-established causal chain in developmental neurotoxicology. The relationship would be:

- **Biological plausibility:** Very high. This is essentially the mechanism underlying many hypomyelination conditions (e.g., Pelizaeus-Merzbacher disease, some thyroid hormone disruption phenotypes).
- **Essentiality:** Reducing oligodendrocyte differentiation experimentally (e.g., via Notch pathway manipulation or thyroid hormone antagonism) reproducibly causes hypomyelination in animal models.
- **Currently missing:** KE2107's only listed upstream is KE2223 ("Reduced number of oligodendrocytes"). KE2116 ("Decreased differentiation") is a conceptually earlier and complementary event — impaired differentiation logically precedes and contributes to

reduced oligodendrocyte numbers. Both pathways lead to hypomyelination and should be captured.

Recommendation: Propose a new KER: **KE2116 → KE2107** (cell → tissue level, within-organism, developmental).

2. KE2116 → KE2223 (New KER: Decreased differentiation → Reduced number of oligodendrocytes)

This would also be a valid and arguably necessary relationship. KE2223 ("Reduced number of oligodendrocytes") currently sits upstream of KE2107, but its own upstream connections are limited to KE1656 (Thyroid Receptor Antagonism) and KE2115 (Altered cholesterol metabolism). KE2116 is a more proximal and direct cause of reduced oligodendrocyte numbers — fewer cells differentiating means fewer mature cells. This would strengthen the existing chain:

KE2116 → KE2223 → KE2107

3. KE2107 → KE1977 (New KER: Hypomyelination → Disruption of sodium channel gating kinetics)

This is biologically plausible but more complex, and represents a cross-pathway connection worth discussing carefully.

The biology: Myelin sheaths do not merely insulate axons — they actively organize the spatial clustering and composition of voltage-gated sodium channels (VGSCs) at nodes of Ranvier. Specifically:

- Node formation (clustering of Nav1.6 at nodes) depends on intact myelination and paranodal axo-glial junctions.
- In hypomyelinated axons, node geometry is disrupted, VGSC clustering is diffuse or lost, and the kinetic properties of sodium channels are altered because their local membrane environment and associated proteins (Ankyrin-G, contactin, neurofascin) are disorganized.
- This means hypomyelination can directly cause altered gating kinetics — the biology underpins KE2107 → KE1977.

Assessment: This KER would be a tissue-to-cell level "backward" connection in terms of organization, but it is directionally correct temporally (myelination precedes node maturation). It is less commonly captured in AOPs but represents an important mechanistic link. The evidence base for this in developmental contexts (e.g., shiverer mice, thyroid-deficient animals) supports it.

Can Existing Common Relationships and Upstream KEs Bridge These Events?

Yes — there is a shared upstream convergence point. Examining the network:

KE1656 (TR Antagonism) —→ KE2223 (Reduced oligodendrocytes) —→ KE2107 (Hypomyelination)
KE2115 (Cholesterol metabolism) —→ KE2223

KE1353 (VGSC Binding) —→ KE1977 (Disrupted Na channel kinetics) —→ KE1983 (Disrupted AP generation)

KE2116 currently floats as an orphan, but it logically belongs between the thyroid/cholesterol upstream events and KE2223. Once KE2116 is connected to KE2223 and KE2107, the myelination arm of the network becomes a proper linear chain feeding into the sodium channel / action potential arm via the proposed KE2107 → KE1977 KER.

The existing upstream for KE1977 already includes **KE1353** (VGSC binding as MIE in AOP442 and AOP489), which is a chemically-initiated event. The myelination-driven disruption of VGSC kinetics would represent a *non-MIE, developmental* route to the same cellular KE — this is scientifically important and would make KE1977 a convergence node for two distinct upstream pathways (direct channel binding vs. structural myelination failure).

Summary of Proposed New KERs

Proposed KER	Direction	Level Transition	Plausibility	Priority
KE2116 → KE2223	Differentiation → Cell number	Cell → Cell	High	High
KE2116 → KE2107	Differentiation → Tissue myelination	Cell → Tissue	High	High
KE2107 → KE1977	Hypomyelination → Na channel kinetics	Tissue → Cell	Moderate-High	Medium

The first two KERs are low-hanging fruit with strong existing literature support. The third is more novel but mechanistically grounded in node of Ranvier biology and would be a meaningful contribution to the AOP-Wiki network, particularly for developmental neurotoxicity pathways involving thyroid disruption or cholesterol metabolism perturbation.